

Real Estate Machine

1. Dataset

- Dataset: [House Data \(Kaggle\)](#)
- Size: ~4600 records, 21 columns.

Features:

- `id` → Unique identifier for each house
- `date` → Date of house sale
- `price` → Sale price (target for regression)
- `bedrooms`, `bathrooms` → Number of rooms
- `sqft_living`, `sqft_lot` → Living space and lot size
- `floors` → Number of floors
- `waterfront` → Whether house has waterfront view (binary)
- `view` → Quality of view (0-4)
- `condition` → House condition (1-5)
- `grade` → Construction/architecture grade (1-13)
- `sqft_above`, `sqft_basement` → Above-ground and basement area
- `yr_built`, `yr_renovated` → Year built / renovated
- `zipcode` → Area code
- `lat`, `long` → Geographic coordinates
- `sqft_living15`, `sqft_lot15` → Adjusted living/lot area (2015 data correction)

2. Project Pipeline

Students must follow this pipeline for their project:

1. Data Cleaning & Understanding

- Handle missing values, outliers, and data types.
- Explore distributions, correlations, and feature importance.

2. Exploratory Analysis & Business Insights

- Analyze price drivers
- Provide recommendations for a real estate company (e.g., where to invest, which features matter most).

3. Unsupervised Learning (Clustering)

- Apply clustering to group properties.
- Define meaningful categories for clusters.

4. Category Classification

- Preprocess data and build a classifier that predicts the cluster category for new houses.

5. Price Regression Modeling

- Build regression models to predict prices.
- Evaluate using metrics.
- Students must research market-acceptable error levels.

6. LLM-based Model Interpretation

- Use a Large Language Model (LLM) to interpret regression predictions:
 - Why was this price predicted?
 - Is it realistic compared to the market?

7. Deployment

- Build a Streamlit app or API + web application:
 - Input house details.
 - Classify house into a category.
 - Predict price.
 - LLM interprets the result.
 - Display outputs in a user-friendly interface.

8. Business Presentation

- Create a professional presentation:
 - Dataset overview.
 - Insights & recommendations.
 - Model results.
 - Demo of deployment.

3. Deliverables

- Analysis Code: EDA + preprocessing.
- Modeling Code: Classification + regression and Clustering.
- Saved Models: Serialized models (.pkl/.joblib).
- Business Presentation: PowerPoint/Slides.
- Kaggle Notebook Link with the full project.

4. Evaluation Criteria

- **Analytical Thinking:** Problem framing and business understanding.
 - **Data Processing:** Correct cleaning and preprocessing choices.
 - **Modeling Accuracy:** Proper evaluation of models, with justified metrics.
 - **Deployment:** Streamlit or Working web app/API with integrated pipeline.
 - **Presentation Skills:** Clarity, structure, and business orientation.
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⚡ *Reminder: The project is not only about building models but also about demonstrating end-to-end thinking: from raw data
→ analysis → clustering → supervised learning → regression
→ interpretation → deployment → business presentation.*

AND THANK YOU FOR YOUR EFFORT 😊